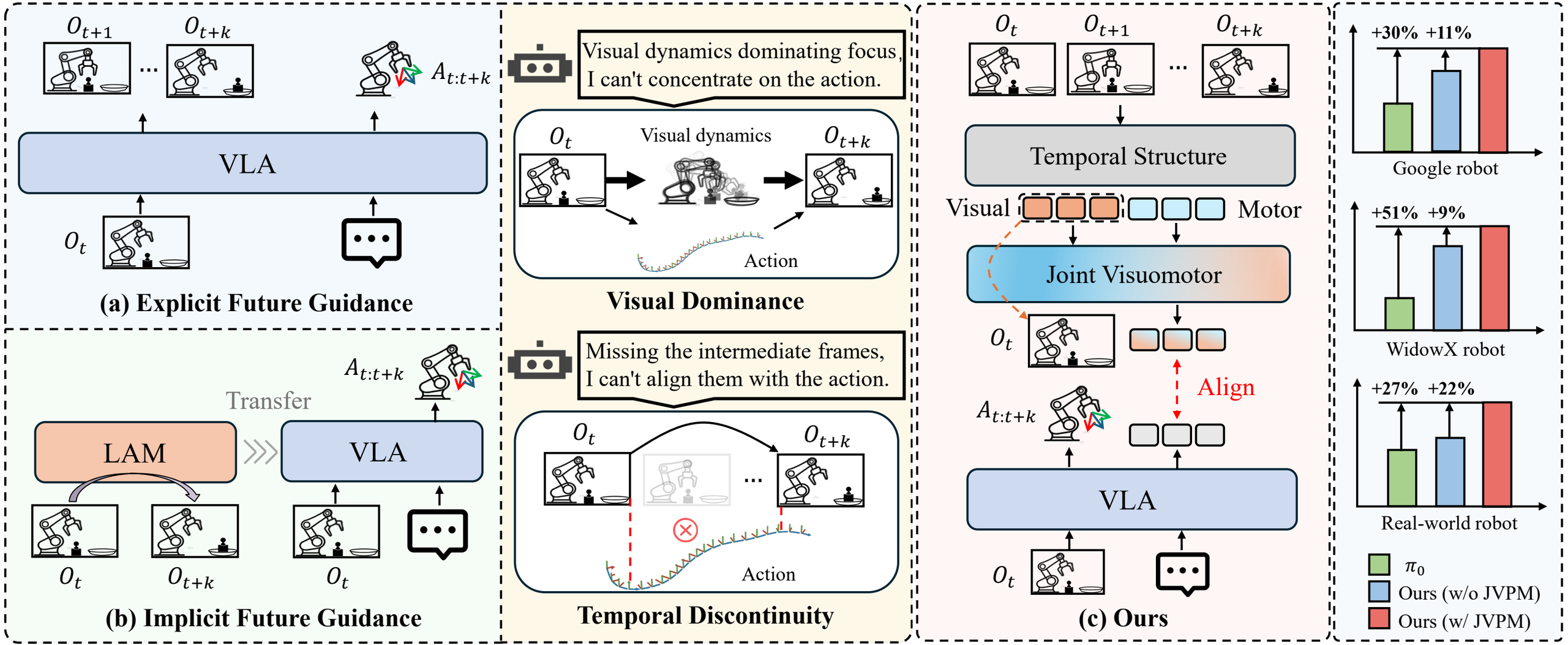
O_{t+1}
O_{t+k}
$A_{t:t+k}$
VLA
O_t
(a) Explicit Future Guidance
$A_{t:t+k}$
Transfer
LAM
VLA
O_t
O_{t+k}
O_t
(b) Implicit Future Guidance
Visual dynamics dominating focus, I can't concentrate on the action.
O_t
Visual dynamics
O_{t+k}
Action
Visual Dominance
Missing the intermediate frames, I can't align them with the action.
O_t
O_{t+k}
Action
Temporal Discontinuity
O_t
O_{t+1}
O_{t+k}
Temporal Structure
Visual
Motor
Joint Visuomotor
O_t
Align
$A_{t:t+k}$
VLA
O_t
(c) Ours
+30% +11%
Google robot
+51% +9%
WidowX robot
+27% +22%
Real-world robot
π_0
Ours (w/o JVPM)
Ours (w/ JVPM)